LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a fraction by a fraction means, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Rivera is dividing a 3/4-pound evidence bag into portions that each weigh 1/8 pound. She needs to know exactly how many portions she can make for testing. Can you solve the case?

REMEMBER

Division Expressions with Fractions and Mixed Numbers

Dividing Fractions by Fractions. Formula: a/b ÷ c/d = a/b × d/c When you divide a fraction by a fraction, you keep the first fraction, change ÷ to ×, and flip the second fraction (the divisor) to its reciprocal. Then multiply and simplify.

1. Name the question: 3/4 ÷ 1/8 asks how many 1/8s fit in 3/4.
2. Use common denominators: 3/4 = 6/8, and 6 eighths ÷ 1 eighth =
3. Shortcut: multiply by the reciprocal of the divisor, 3/4 × 8/1 = 24/4 = 6, then check by multiplying: 6 × 1/8 = 6/8 = 3/4

Words: **Dividend *(Dividendo)*** **Divisor *(Divisor)*** **Reciprocal *(Recíproco)*** **Quotient *(Cociente)*** **Simplify *(Simplificar)***

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 6 ÷ 1/2?

(A) 1/12 (B) 3

(C) 6 1/2 (D) 12

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 6 ÷ 1/3?

(A) 2
(B) 3
(C) 6/3
(D) 18

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3 ÷ 1/5?

(A) 3/5
(B) 5/3
(C) 8
(D) 15

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 10 ÷ 1/4?

(A) 2.5
(B) 10/4
(C) 14
(D) 40

SHOW YOUR WORK

PART 2 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

The lab printer fills 2/5 of a box of evidence labels in 1/2 hour. At that rate, how much of a box does it fill in ONE hour?

(A) 1/5 box

(B) 1 1/4 boxes

(C) 4/5 box

(D) 9/10 box

HINT How many half-hours are in one hour?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** 1/2 ÷ 1/4
2. **Divide straight across** (1 ÷ 1) / (2 ÷ 4) = 1/(1/2)
3. **Answer** = 2

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT You cannot divide fractions by dividing the numerators and denominators straight across.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

SENTENCE STARTER 5/6 ÷ 1/12 = 10 because 5/6 is the same as ___ twelfths, and ___ twelfths divided by 1 twelfth = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is (3/4) ÷ (1/2)?

(A) 1/2 (B) 2/3

(C) 3/2 (D) 3/8

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

2 Circle the letter of the best answer. Show how you know.

What is (2/3) ÷ (4/9)?

(A) 2/3
(B) 3/2
(C) 8/27
(D) 9/4

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

3 Circle the letter of the best answer. Show how you know.

What is (5/6) ÷ (5/12)?

(A) 1/2
(B) 2
(C) 12/5
(D) 25/72

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

PART 2 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

A carpenter has a board that is 5/6 of a foot long. She needs to cut pieces that are each 1/12 of a foot. She wants to know how many pieces she can cut from the board. 5/6 ÷ 1/12 = ?

(A) 2
(B) 5/72
(C) 10
(D) 12

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A carpenter has a board that is 5/6 of a foot long. She needs to cut pieces that are each 1/12 of a foot. She wants to know how many pieces she can cut from the board. Rewrite 5/6 with a denominator of 12.

(A) 5/12
(B) 6/12
(C) 10/12
(D) 11/12

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

6 Circle the letter of the best answer. Show how you know.

A carpenter has a board that is 5/6 of a foot long. She needs to cut pieces that are each 1/12 of a foot. She wants to know how many pieces she can cut from the board. If each piece were 1/6 of a foot, how many pieces?

(A) 5
(B) 6
(C) 10
(D) 30

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help